Energy in Food Experiment

Apparatus and Chemicals

Boiling tube,
1Clamp stand,
Boss Head and clamp,
1Bunsen burner,
Heatproof mat,
1 Measuring cylinder, 10mL,
1 Mounted needle,
Tongs or forceps for food samples that cannot be impaled
Thermometer (–10 °C–110 °C)
Eye protection
Water
Balance, accurate to ± 0.1 g
Food sample

Procedure

- Use a range of foods – around 1 cm square or 0.5 cm cubed.
- Use the measuring cylinder to measure 10 mL of water into the boiling tube.
- Clamp the boiling tube to the retort stand.
- Measure the temperature of the water with the thermometer. Record the temperature in a suitable results table.
- Choose a piece of food and find its mass using the balance. Record the mass in the table.
- Impale the piece of food carefully on a mounted needle.
- Light the Bunsen burner and hold the food in the flame until it catches alight.
- As soon as the food is alight, put it under the boiling tube of water. Try to make sure that as much of the heat from the burning food as possible is transferred to the water. Do this by keeping the flame under the tube. (The food should be about 1cm from the boiling tube.)
- Hold the food in place until the food has burnt completely. If the flame goes out, but the food is not completely burnt, quickly light it again using the Bunsen burner and replace the food beneath the tube.
- As soon as the food has burned away completely and the flame has gone out, measure the temperature of the water again. Before measuring, stir the water carefully with the thermometer and note down the highest temperature reached in the results table.
- Calculate the rise in temperature.
- Calculate the energy released from each food by using this formula.

How Much Energy Is There In Food Equation

We measure the energy in foods in either joules or calories and you can see this displayed on food packets. A joule is the energy required to raise the temperature of 1g of water by 1°C, and a calorie is 4.2 joules.

The formula you need to use is:

$$\text{Energy per gram} = \frac{\text{mass of water (g)} \times 42 \times \text{change in temperature of water}}{\text{Mass of food burned}}$$

4.2 is the value of the specific heat capacity of water, in joules per gram per degree Celsius. In other words it is the number of joules taken to raise the temperature of water by 1 °C. We are using 42 in our calculations as we used 10g of water. Note that 1 mL of water has a mass of 1 g.

If the number is more than 1000 J/g, express it as kilojoules (kJ):
1 kilojoule = 1000 joules

Compare your results for each food with the rest of the class.

Food	Mass of food (g)	Initial temperature (°C)	Final temperature (°C)	Temperature Change (°C)	Energy per gram $Q = \frac{\text{water mass} \times 42 \times \Delta T}{\text{food mass}}$

Questions

1. Which food contained the most energy per gram? _____

2. Define a joule.

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